

Accuracy of a screening instrument to identify potential child abuse in emergency departments^{☆,☆☆}



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ABSTRACT

Although screening for child abuse at emergency departments (EDs) increases the detection rate of potential child abuse, an accurate instrument is lacking. This study was designed to measure the accuracy of a screening instrument for detection of potential child abuse used in EDs. In a prospective cohort study at three Dutch EDs, a 6-item screening instrument for child abuse, *Escape*, was completed for each child visiting the ED. The data from the completed *Escape* instrument was used to calculate sensitivity, specificity, and the positive/negative predictive value per item. The clinical notes and conclusions of the screen instruments of all potentially abused children reported to the hospitals' Child Abuse Teams were collected and reviewed by an expert panel. A logistic regression model was used to evaluate the predictors of potential abuse. Completed *Escape* instruments were available for 18,275 ED visits. Forty-four of the 420 children with a positive screening result, and 11 of the 17,855 children with a negative result were identified as potentially abused. Sensitivity of the *Escape* instrument was 0.80 and specificity was 0.98. Univariate logistic regression showed that potentially abused children were significantly more likely to have had an aberrant answer to at least one of the items, $OR = 189.8$, 95% CI [97.3, 370.4]. Most of the children at high risk for child abuse were detected through screening. The *Escape* instrument is a useful tool for ED staff to support the identification of those at high risk for child abuse.

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Introduction

Early intervention in childhood abuse is important to prevent or reduce long-term adverse effects (Afifi et al., 2007; Chartier, Walker, & Naimark, 2007; Ethier, Lemelin, & Lacharite, 2004; Felitti et al., 1998). Although screening for child abuse at emergency departments (EDs) is known to increase the detection rate of potential child abuse, an accurate screening instrument for use in the ED setting is still lacking (Louwers, Affourtit, Moll, Koning, & Korfage, 2010; Woodman et al., 2008). Because EDs have a high turnover of patients and staff work under considerable pressure and time constraints, a short and reliable screening instrument is needed that can be completed quickly.

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1. Is the history consistent?	Yes	No
2. Was seeking medical help unnecessarily delayed?	Yes	No
3. Does the onset of the injury fit with the developmental level of the child?	Yes/N. A.	No
4. Is the behavior of the child, his or her carers and their interaction appropriate?	Yes	No
5. Are findings of the head-to-toe examination in accordance with the history?	Yes	No
6. Are there other signals that make you doubt the safety of the child or other family members?	Yes*	No
*If Yes describe the signals in the box 'Other comments' below.		
Other comments		

Fig. 1. 'Escape instrument': the screening instrument for child abuse used at the emergency departments. One (or more) ticked answers in the dark boxes indicate the possibility of an increased risk of child abuse and further action is recommended.

A team of pediatricians and screening experts developed an instrument to screen for child abuse in EDs to identify high-risk children. The design was based on a systematic literature review (Louwers et al., 2010), earlier screening instruments (Benger & Pearce, 2002; Bleeker, Vet, Haumann, van Wijk, & Gemke, 2005; Pless, Sibald, Smith, & Russell, 1987; Sidebotham & Pearce, 1997), interviews with professionals, and pretesting of the concept with emergency department nurses (Fig. 1). This screening instrument, *Escape*, is a 6-item checklist addressing risk factors for child abuse, which may be predictive for child abuse in any child. The instrument is to be used irrespective of the patient's reason for their visit; it is not an injury evaluation checklist. The present study was designed to measure the accuracy of this newly developed screening instrument for child abuse in EDs using expert panels. The possibility to minimize the burden of completing the instrument while maintaining sensitivity and specificity was also examined.

Methods

Intervention

The *Escape* instrument was implemented in three Dutch hospitals (Louwers et al., 2012), where it was to be used in each child aged 18 years or younger who visited the ED. ED nurses completed the *Escape* instrument during the triage of the patients. The typical triage included a head-to-toe assessment. The goals of the head-to-toe assessment were to examine the whole skin for rashes, to look at the hydration state, and to ensure no injuries were missed. If one or more items of the instruments were aberrant, the screening result was considered positive. The nurse was instructed to inform the ED physician of the result of the screening.

The physician provided care as usual to every patient and had the final responsibility to evaluate the increased risk of child abuse. When it was very clear that a child had been abused (e.g., a fracture in an infant or signs of inflicted traumatic brain injury) or when the ED physician remained concerned about the safety of the child after taking the history of and examining the child, the physician referred the child to the hospital's Child Abuse Team for further care (irrespective of the screening result). The Child Abuse Teams evaluated every case that was referred to them and assessed which action was needed to increase the safety of the child (e.g., a call to the general practitioner of the child or a referral to Child Abuse Centers). Child Abuse Centers explore the cases of suspected child abuse and take care of adequate aid if necessary. All kinds of professionals and citizens can voluntarily report suspected child abuse to the Child Abuse Centers, there is no mandatory reporting in the Netherlands. It was also possible that the ED physician was reassured that abuse did not occur after taking the history of and examining the child and so did not refer the child to the Child Abuse Team.

Data from all *Escape* instruments completed between July 2008 and December 2009 (18 months) were used to measure the accuracy of this screening instrument for child abuse in EDs. The study was approved by the Medical Ethical Committee of the Erasmus MC (MEC-2007-195).

Case definition

The aim was to ensure that data of all cases of potential child abuse were collected from each ED and to establish to what extent cases were uniformly defined across the participating departments. In doing so, we first contacted the Child Abuse Teams of the three hospitals and collected data on all potentially abused children who had been reported by the ED staff during the study period. Subsequently, to establish whether or not these cases were child abuse, the data were independently evaluated by an expert panel consisting of four physicians with extensive experience in child abuse, of whom one was a forensic pediatrician and two were pediatricians.

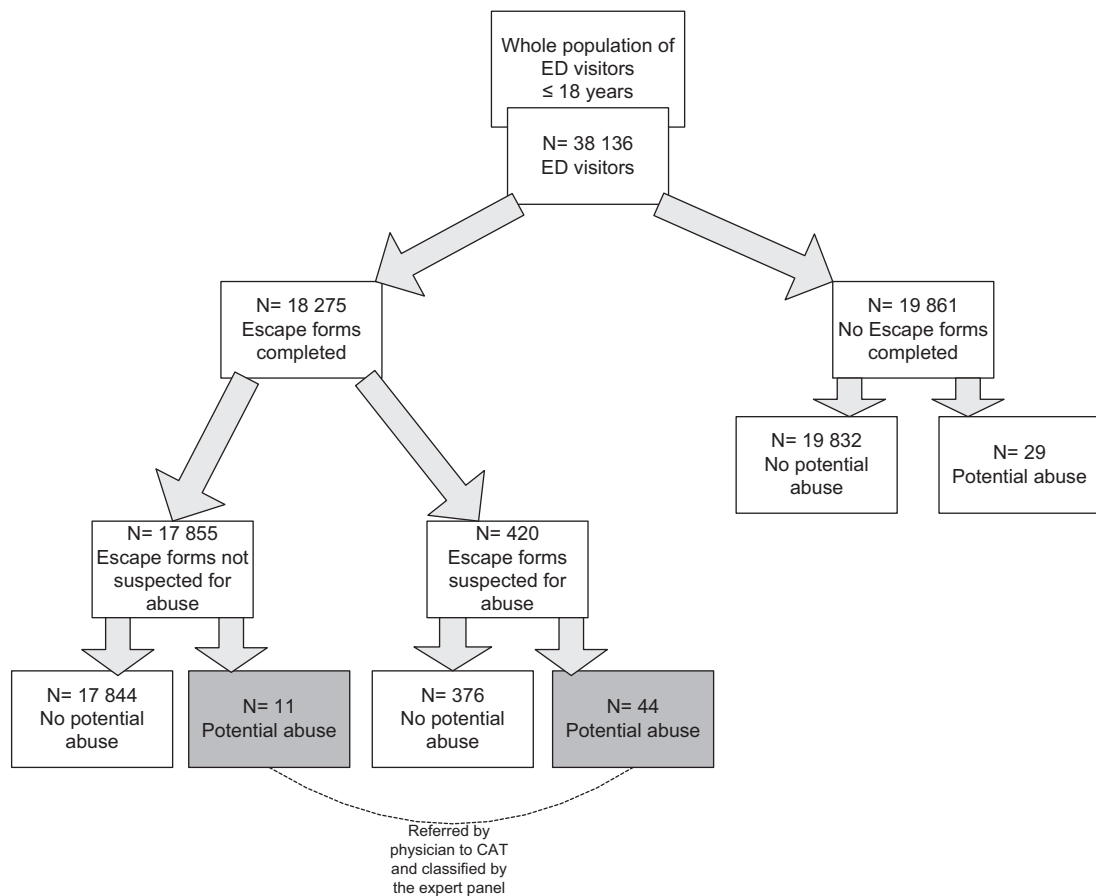


Fig. 2. Flow diagram of screening for child abuse in the three emergency departments.

The evaluators classified the cases based on their review of the clinical notes. The notes included information regarding age, gender, signs at presentation at the emergency department, history and findings at the emergency department, conclusion of the screening instrument, and diagnosis (of the physician).

For this study, eight inclusion criteria and four exclusion criteria were formulated (see [Appendix A](#)) based on the following definition for child abuse: *Any form of threatening or violent physical, mental or sexual interaction with a minor which is perpetrated actively or passively by parents or other persons on whom the minor is dependent and causes or will probably cause physical or mental injury and serious harm to the minor* (*“Wet op de Jeugdzorg,”* 2010). For an individual patient, if a professional indicated one or more of the inclusion criteria to be present, that patient was classified as a *potential case*. If a professional indicated that one or more of the exclusion criteria was present, that patient was excluded from the list of potential cases. A patient was also excluded from the list if both inclusion and exclusion criteria were indicated. For the analyses, we considered patients as potential cases if at least two or more professionals classified them as such (see [Louwers et al., 2012](#), for details).

Because the aim was to measure the accuracy of screening, we excluded children who were known to have been abused at the moment they visited the emergency department (e.g., the carers/patients reported it themselves; they were brought in by the police). To prevent an overestimation of the results of screening for child abuse we also excluded cases of alcohol intoxication and suicide attempts, because the reasons for their ED visits are often already known at presentation at the ED (ambulance personnel or family members can clarify the situation most of the times).

Statistical analysis

Chi-squared tests were used to compare categorical variables of children who classified as potential cases of child abuse and those who did not. To validate the Escape instrument's sensitivity, specificity, and positive and negative predictive values, likelihood ratios were calculated. Both univariate and multivariate regression analyses were used to determine the predictive value of each single item.

Table 1

Characteristics of all emergency department (ED) visitors (aged ≤ 18 years) in three Dutch hospitals classified as whether or not screened for child abuse with the Escape instrument and whether or not potentially abused.

Characteristics	Children screened for child abuse with Escape instrument			Children not screened for child abuse with Escape instrument		
	Potential child abuse <i>n</i> = 55	No potential child abuse <i>n</i> = 18,220	<i>p</i> -Value ^a	Potential child abuse <i>n</i> = 29	No potential child abuse <i>n</i> = 19,832	<i>p</i> -Value ^a
Age			0.002			0.08
0–4 years	44 (80%)	9,991 (55%)		18 (62%)	9,741 (49%)	
5–8 years	4 (7%)	3,533 (19%)		6 (21%)	3,451 (17%)	
9–12 years	2 (4%)	2,612 (14%)		5 (17%)	2,903 (15%)	
13–18 years	5 (9%)	2,084 (11%)		0	3,737 (19%)	
Sex (male)	24 (44%)	10,298 (57%)	0.054	11 (38%)	11,378 (57%)	0.03
Referrer			<0.001			0.22
Self-referral	31 (56%)	9,532 (52%)		16 (55%)	10,147 (51%)	
Ambulance	12 (22%)	1,060 (6%)		5 (17%)	1,188 (6%)	
General practitioner	7 (13%)	4,013 (22%)		4 (14%)	4,522 (23%)	
Other	3 (6%)	3,166 (17%)		3 (10%)	3,461 (17%)	
Missing	2 (4%)	449 (3%)		1 (3%)	514 (3%)	
Treating specialist			<0.001			0.15
Surgeon	28 (51%)	5,771 (32%)		16 (55%)	6,539 (33%)	
Pediatrician	19 (35%)	10,934 (60%)		11 (38%)	11,069 (56%)	
Other	8 (14%)	1,353 (7%)		2 (7%)	2,162 (11%)	
Missing	0	162 (1%)		0	62 (0.3%)	
Destination after ED visit			<0.001			0.001
Home without control	16 (29%)	9,749 (54%)		8 (28%)	9,396 (47%)	
Hospital admission	16 (29%)	2,520 (14%)		5 (17%)	3,151 (16%)	
Outpatient clinic	12 (22%)	2,678 (15%)		6 (21%)	3,157 (16%)	
Other	7 (13%)	1,847 (10%)		8 (28%)	1,380 (7%)	
Missing	4 (7%)	985 (5%)		1 (3%)	1,371 (7%)	
General practitioner	0	441 (2%)		1 (3%)	1,377 (7%)	
Screening positive (≥ 1 item positive)	44 (80%)	376 (2%)	<0.001	0	0	NA
Number of positive items			<0.001			NA
≥ 2	22 (40%)	86 (0.5%)		0	0	
≥ 4	6 (11%)	19 (0.1%)		0	0	

^a Calculated with the Chi-square test. NA = not applicable.

To examine the possibility of limiting the number of items in the instrument, the least sensitive items were removed and the sensitivity and specificity analyses were performed again. Statistical significance was defined as $p < .05$. The SPSS version 17.0 was used for analysis.

Results

During the 18-month study period, the three EDs were visited by a total of 38,136 children aged 18 years and younger. These children were on average 5.5 years old, 57% were male, 52% presented without being referred, 58% were treated by a pediatrician, and 32% had a surgical problem.

For 18,275 (48%) of these children, the Escape instrument was completed (see Fig. 2). Overall, 2.3% ($n = 420$) of the instruments were positive. The responsible physician in the ED referred 89 screened patients (positive and negative) to the Child Abuse Teams. Subsequently, of these 89 patients, the expert panel classified 55 (56%) of them as cases of potential abuse. Of these 55 cases, 44 (80%) were positive (Table 1). In 62% of the 89 cases, three or more panel members agreed about whether there was potential abuse.

In 19,861 of the ED visits (52%), no Escape instruments were completed (Table 1). Although the emergency department nurses were urged to complete the Escape instrument, it was not a mandatory part of the electronic patient file, and therefore, was not always completed. As shown in Table 1, the nonscreened group was somewhat older than the screened group, and they were referred more often for further care after the ED visit.

Table 2 presents the sensitivity, specificity, and positive/negative predictive values of (items of) the Escape instrument. The highest sensitivity of a single item was 0.59 (30/51) and the lowest was 0.12 (6/52). The specificity for each single item was 0.99. Sensitivity of the Escape instrument (≥ 1 item positive) was 0.80 (44/55) and specificity was 0.98 (17,844/18,220).

Table 2 also presents positive and negative predictive values. The positive likelihood ratio of the Escape instrument was 40, and the negative likelihood ratio was 0.20.

When item 2 (with the lowest sensitivity) was excluded, the sensitivity of the instrument decreased from 0.80 to 0.73. However, when excluding items 1, 4, or 5, the sensitivity decreased to 0.78, thus resulting in one missed case per excluded item. If items 1, 4, and 5 had been excluded, the sensitivity would decrease to 0.75, the specificity would remain at 0.98, and three cases would have been missed.

Table 2

Univariate comparisons of predictor variables in the Escape instrument for child abuse in emergency departments between cases of suspected child abuse and the total pediatric population, and sensitivity, specificity, positive and negative predictive value.

Items	Aberrant answer	No. of aberrant answers in total population <i>n</i> = 18,275	No. of aberrant answers among cases of suspected abuse <i>n</i> = 55	Missings	Odds ratio (95% CI)	<i>p</i> -Value	Sensitivity (95% CI)	Specificity (95% CI) ^b	Positive predictive value (95% CI)	Negative predictive value (95% CI) ^c
1. Is the history consistent?	No	83 (0.5%)	9 (16%)	53	50.0 (23.6–106.2)	<0.001	0.17 (0.09–0.30)	0.99	0.11 (0.05–0.20)	0.99
2. Was seeking medical help unnecessarily delayed?	Yes	141 (0.8%)	6 (11%)	130	17.4 (7.3–41.3)	<0.001	0.12 (0.05–0.24)	0.99	0.04 (0.02–0.09)	0.99
3. Does the onset of the injury fit with the developmental level of the child?	No	81 (0.4%)	17 (31%)	1,136 ^a	137.0 (72.7–258.5)	<0.001	0.34 (0.22–0.49)	0.99	0.21 (0.13–0.32)	0.99
4. Is the behavior of the child, his or her carers and their interaction appropriate?	No	85 (0.5%)	11 (20%)	152	65.3 (32.3–131.9)	<0.001	0.21 (0.12–0.35)	0.99	0.13 (0.07–0.22)	0.99
5. Are findings of the head-to-toe examination in accordance with the history?	No	54 (0.3%)	9 (16%)	109	82.1 (37.9–178.2)	<0.001	0.17 (0.09–0.30)	0.99	0.17 (0.08–0.30)	0.99
6. Are there other signals that make you doubt the safety of the child or other family members?	Yes	170 (0.9%)	30 (55%)	152	182.9 (102.3–327.4)	<0.001	0.59 (0.44–0.72)	0.99	0.18 (0.12–0.24)	0.99
≥ 1 question positive		420 (2.3%)	44 (80%)	0	189.8 (97.3–370.4)	<0.001	0.80 (0.67–0.89)	0.98	0.10 (0.08–0.14)	0.99

^a Nurses skipped this question in case of a trauma when the option 'not applicable' was not yet available.

^b Maximum range of 95% CI 0.98–0.99.

^c Maximum range of 95% CI 0.997–0.999.

Because of the extremely high correlations between the predictors, multivariate analysis of the six items of the Escape instrument added no new information about the validity of the instrument.

Discussion

The Escape instrument proved to be useful to support ED staff in identifying the group of children at high risk of potential child abuse. Using univariate logistic regression to measure the accuracy of the instrument, it was found that cases of potential child abuse were significantly more likely to have had an aberrant answer on at least one of the items (and thus to be screen-positive) compared with the total population.

Although the sensitivity and positive predictive values of each single item were moderate, sensitivity for the complete Escape instrument was 0.80, which indicates that not all potential cases of child abuse were detected when using a positive Escape instrument. However, specificity and the negative predictive value of each item were high, indicating that child abuse was not likely when the Escape instrument was negative.

Each year, approximately 3% of Dutch children are victims of child abuse (Euser, van Ijzendoorn, Prinzie, & Bakermans-Kranenburg, 2010). In the present cohort, in only 0.3% of the screened ED patients and 0.1% of children not screened was potential abuse detected. For both situations, this percentage is very low. This could be related to the fact that most detected cases at the EDs concern physical abuse, while the 3% rate as reported by Euser and colleagues consists all types of child abuse including neglect and emotional abuse.

To increase the detection of child abuse in EDs adherence to the screening protocol needs to increase. Therefore it is advisable to minimize the time and effort it takes to conduct screening, while EDs are generally very busy with diverse patients and staff working under considerable pressure and time constraints. The number of items of the screening instrument should be as limited as possible without decreasing its reliability.

Including item 1, 4, or 5 resulted in each detecting one additional case. Item 1, about a consistent history, and item 4, about the interaction between the carers and the child or the behavior of the child, apparently did not result in detecting many cases, but on the other hand, completing these items does not take much time either. Completing item 5, concerning the head-to-toe examination of the patient by the nurse during the triage process to detect signs of child abuse, does take time. The sensitivity of this item was unexpectedly low. This finding could be related to two possible mechanisms: the head-to-toe examination was not properly completed or the examination did result in only minimal additional sensitivity. In the latter case, the finding might imply that conducting the head-to-toe examination is not worth the effort. When considering the balance between the time/effort made by the staff to motivate the patients and/or their carers to undergo the head-to-toe examination versus the limited contribution to the detection rate, exclusion of this item from the Escape instrument seems feasible. On the other hand, because no cases of child abuse should be missed, it might be best to err on the side of inclusion. Further study on the effectiveness of the head-to-toe examination is warranted.

In 11 patients potential child abuse was signified by ED staff, while the corresponding Escape instruments were negative. This finding suggests that mere implementation of the instrument is insufficient to achieve optimal effects of screening (Louwers et al., 2012). These false negative cases of potential abuse also show that ED nurses need to be trained in recognizing the risk factors/signals for child abuse and in communicating with the parents/child when they suspect child abuse. Physicians may also need training in how best to recognize, handle, and communicate potential child abuse.

Identifying the risk of potential child abuse and the need for an effective intervention to reduce this risk are inextricably linked (Raffle & Gray, 2007). If a physician is concerned about the safety of a child, the hospital is responsible to provide the facilities necessary for further research, therapy, and follow-up of that child. The hospital can consult Child Protective Services, who have the facilities to examine potential child abuse and the expertise to refer the child to adequate care and to ensure that the child develops in a safe environment.

One limitation of this study is the fact that the rate of confirmed child abuse was unavailable and that using data of potential cases could result in an overestimation of the rate of true cases of child abuse. Follow-up of the potential cases is recommended to confirm the accuracy of the Escape instrument.

Strengths of the study are the multicenter setting, the large number of completed Escape instruments, and the number of potential cases identified.

Conclusions

In this study, the majority of children at high risk for child abuse were identified by screening at EDs. The Escape instrument is useful for emergency department staff to identify the group of children at high risk for potential child abuse.

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Appendix A.

Inclusion and exclusion criteria for the cases of this study.

Inclusion criteria

1. Injury caused by a person on whom the child is dependent
2. Injury resulted from neglect by caregivers
3. Psychological harm may have resulted from actions of the person on whom the child is dependent
4. Psychological harm may have resulted from failure of the person on whom the child is dependent
5. Withheld from medical care
6. Child witnessed domestic violence
7. Child witnessed sexual acts
8. Child was victim of sexual acts

Exclusion criteria

1. Suspicion of abuse was known prior to emergency department visit
2. Alcohol intoxication
3. Suicide attempt
4. Injury caused by a stranger or by peers